WORKED SOLUTION

STEP III 2007, Question 4

Pure • Parametric Curves and Curvature

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q4

The key idea

Everything rests on a clean formula for $\dot{x}$. The awkward term $\ln\tan\frac{1}{2}t$ differentiates to $1/\sin t$, and then $\dot{x}$ collapses to $a\cos^2 t/\sin t$. Once we have $\dot{x}$ and $\dot{y}$, the gradient, the tangent, the curvature and the normal all follow by straightforward coordinate geometry. Writing the unit tangent as $(\cos t, \sin t)$ keeps the final part very short.

Solution

The gradient

Since $\dfrac{\mathrm{d}}{\mathrm{d}t}\ln\tan\frac{1}{2}t = \dfrac{\frac{1}{2}\sec^2\frac{1}{2}t}{\tan\frac{1}{2}t} = \dfrac{1}{2\sin\frac{1}{2}t\cos\frac{1}{2}t} = \dfrac{1}{\sin t}$, we get

$$\dot{x} = a\left(-\sin t + \frac{1}{\sin t}\right) = a\,\frac{1-\sin^2 t}{\sin t} = \frac{a\cos^2 t}{\sin t},$$
$$\dot{y} = a\cos t.$$

Both are positive for $0 < t < \frac{1}{2}\pi$, so

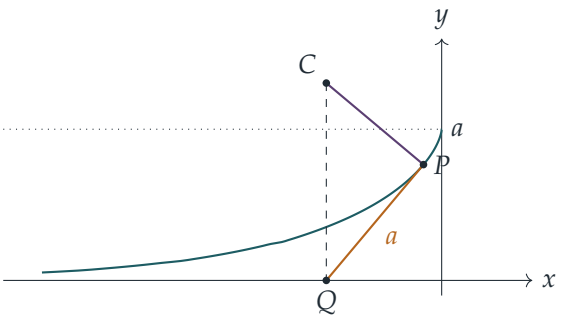
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dot{y}}{\dot{x}} = a\cos t\cdot\frac{\sin t}{a\cos^2 t} = \tan t,$$

as required.

The sketch

- As $t \to 0^+$, $\tan\frac{1}{2}t \to 0^+$, so $x \to -\infty$ while $y \to 0$. The x-axis is an asymptote.
- As $t \to \frac{1}{2}\pi^-$, $x \to a(0+\ln 1) = 0$ and $y \to a$. The curve ends at $(0, a)$, where the gradient $\tan t \to \infty$, so the tangent there is vertical.
- Since $\dot{x} > 0$, x increases with t, and so does the gradient $\tan t$. So the curve rises steadily and is concave upwards, with y between 0 and a.

This curve is the *tractrix*. The figure also shows the points P, Q and C from the later parts.



The tangent: $PQ = a$

The tangent at $P = (x, y)$ has gradient $\tan t$. It meets $y = 0$ after a horizontal run of $y/\tan t$ back from P, so

$$x_Q = x - \frac{y}{\tan t} = x - \frac{a \sin t \cos t}{\sin t} = x - a\cos t.$$

Hence $\overrightarrow{QP} = (a\cos t,\ a\sin t)$, and

$$PQ = \sqrt{a^2\cos^2 t + a^2 \sin^2 t} = a.$$

So the length of the tangent from the curve to the x-axis is always a. (For later use, $x_Q = a \ln \tan \frac{1}{2}t$.)

The radius of curvature

Write $\dot{x} = a\Big(\dfrac{1}{\sin t} - \sin t\Big)$ to differentiate it easily:

$$\ddot{x} = a\Big(-\frac{\cos t}{\sin^2 t} - \cos t\Big), \qquad \ddot{y} = -a\sin t.$$

Then

$$\begin{aligned}\dot{x}\ddot{y} - \dot{y}\ddot{x} &= \frac{a\cos^2 t}{\sin t}(-a\sin t) + a\cos t \cdot a\Big(\frac{\cos t}{\sin^2 t} + \cos t\Big) \\ &= a^2\Big(-\cos^2 t + \frac{\cos^2 t}{\sin^2 t} + \cos^2 t\Big) = a^2\cot^2 t,\end{aligned}$$

and

$$\dot{x}^2 + \dot{y}^2 = a^2\cos^2 t\Big(\frac{\cos^2 t}{\sin^2 t} + 1\Big) = \frac{a^2\cos^2 t}{\sin^2 t} = a^2\cot^2 t.$$

Since $\cot t > 0$ on our range, $(\dot{x}^2 + \dot{y}^2)^{3/2} = a^3\cot^3 t$, and therefore

$$\rho = \frac{a^3\cot^3 t}{a^2\cot^2 t} = a\cot t,$$

as required.

The centre of curvature: CQ is vertical

The tangent at P points along the unit vector $(\cos t, \sin t)$, so the normal points along $(-\sin t, \cos t)$ or its negative.

Ask yourself Which of the two normal directions is "above the curve"?

The vector $(-\sin t, \cos t)$ has positive y-component, since $\cos t > 0$. Moving a short way along it from P takes us above the curve, because the curve is concave upwards (and in any case the other direction takes us downwards). So

$$\overrightarrow{OC} = \overrightarrow{OP} + \rho(-\sin t, \cos t) = \big(x - a\cot t \sin t,\ a\sin t + a\cot t\cos t\big).$$

The x-coordinate is

$$x_C = x - a\cos t = x_Q$$

by the tangent part. So C and Q have the same x-coordinate, and CQ is parallel to the y-axis, as required. (The y-coordinate simplifies to $a\sin t + \dfrac{a\cos^2 t}{\sin t} = \dfrac{a}{\sin t}$, so $C = \big(a\ln\tan\tfrac{1}{2}t,\ a\operatorname{cosec} t\big)$.)

Checks

- With $a = 1$ and $t = \frac{5\pi}{18}$ $(50°)$, $P \approx (-0.1202, 0.7660)$ and $Q \approx (-0.7630, 0)$, giving $PQ = 1.0000$. The formula gives $C \approx (-0.7630, 1.3054)$, directly above Q.
- A numerical check over 2000 random values of a and t, with $\dot{x}$, $\ddot{x}$ found by finite differences, agreed with $\mathrm{d}y/\mathrm{d}x = \tan t$, $PQ = a$, $\rho = a\cot t$ and $x_C = x_Q$ to within rounding error.
- As $t \to \frac{1}{2}\pi$, $\rho \to 0$: the curve bends ever more sharply as it approaches $(0, a)$ (its mirror image in the y-axis meets it there in a cusp). As $t \to 0$, $\rho \to \infty$, matching the curve flattening onto its asymptote.

Common slips

- Differentiating $\ln\tan\frac{1}{2}t$ and forgetting the factor $\frac{1}{2}$ from the chain rule. The answer must come out as $1/\sin t$.
- Leaving $\dot{x}$ as $a(-\sin t + \operatorname{cosec} t)$ without simplifying. The form $a\cos^2 t/\sin t$ is what makes every later part tidy.
- Using the normal direction $(\sin t, -\cos t)$, which points below the curve and puts C in the wrong place.

Beyond STEP

The tractrix is the path traced by a weight dragged on a string of length a by someone walking along the x-axis, and $PQ = a$ is exactly that property. The point C traces the *evolute* of the curve, the locus of its centres of curvature. Here $C = (a \ln \tan \frac{1}{2}t,\ a \operatorname{cosec} t)$ turns out to lie on a catenary. Rotating the tractrix about its asymptote gives the pseudosphere, a surface of constant negative curvature and a model of hyperbolic geometry. Needham's *Visual Complex Analysis* has a beautiful geometric account of the pseudosphere.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.